

Community detection in geometric graphs: Recent advances until 2026

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Résumé **TODO**

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1 Introduction

les titres des paragraphes partiront, c'est pour aider à structurer le rapport
Les parties, la structure, les idées, les références ne sont pas du tout fixes. Ce sont des propositions.

Motivation théorique (RGG) Def informelle : Detection de communautés + spatial graph Expliquer ce qu'est une communauté. Cas particulier du graph clustering (ensembles de points grandement interconnectés et faiblement (dans une certaine mesure)connecté intra-communauté)

Origine : [Penrose, 2003] [Gilbert, 1961]

Motivation géométrique (comparé à SBM) SBM généralisation de Erdos-Renyi [Erdős et al., 1960] Propriétés issues des réseaux : sparse [Del Genio et al., 2011], la propriété "small world", transitivity (friend of a friend is a friend) [Watts and Strogatz, 1998]

Survey : [Abbe, 2018]

Poubelle (hypergraph & attribut) Out of scope + si la transitivity est intéressante alors pourquoi pas hyper-graph [Ruggeri et al., 2023] [Kovács et al., 2025] graphe d'attribut [Liu et al., 2022]

Motivation par application réelle Detection de communautés en général [Fortunato, 2010].

Très bref, parler des réseaux sociaux, embedding word (IA), biologie, wireless network, physique des matériaux (?), propagation d'information (?) [Sankaranarayanan et al., 2020] [Higham et al., 2008] [Gomez et al., 2015]

Problématiques Définition et interprétation, récupération à permutation près

- Distinguabilité [Sankararaman and Baccelli, 2018]
- Récupération partielle (weak recovery)
- Récupération presque exacte (almost exact recovery)
- Récupération exacte (exact recovery) [Gaudio et al., 2024a]

Détection de géométrie (et pas de communauté) Detection problem que l'on abordera pas plus en détail [Bubeck et al., 2016] [Liu et al., 2022] [Mao et al., 2026]

Reconstruction de la géométrie chose qu'on ne fera pas non plus [Valdivia and De Castro, 2019] [Eldan et al., 2022]

2 Modèle

- pas de boucle, pas dirige, pas de communautés chevauchantes, etc...
- espace latent (ou pas) [Castro et al., 2020]
- sphère, espace euclidien?
- régime : dense, logarithmique, sparse
- hard-RGG, soft-RGG [Duchemin and De Castro, 2023]
- motivation : bien que les modèles ne soient pas les mêmes, ils n'ont pas pour but de comprendre tous ceux possible mais qu'à travers chaque classe voir les mécanismes. Exemple : passer du local au global puis raffinement, les quantités utilisées, ...

Le but est de faire des notations claires, concises et générales pour pouvoir expliquer tous les modèles sans avoir à se perdre dans la notation.

3 Seuil théorique information

Faire le rapport dans l'ordre chronologique je pense

- [Galhotra et al., 2018] (Geometric Block Model, GBM)
- [Sankararaman and Baccelli, 2018] & [Abbe et al., 2021] (Geometric SBM)
- [Chien et al., 2020]
- [Galhotra et al., 2020] (GBM)
- [Galhotra et al., 2023] (GBM)
- [Avrachenkov et al., 2024] (Geometric Kernel Block Model, GKBM)
- [Gaudio et al., 2024b] (Geometric SBM)
- [Gaudio et al., 2024a] (Geometric Hidden Community Model, GHCM)
- [Gaudio and Guan, 2025] (GHCM)
- [Gaudio and Jin, 2026] (GHCM)

3.1 Geometric Block Model (GBM)

[Galhotra et al., 2018] [Galhotra et al., 2020] [Galhotra et al., 2023]
[Chien et al., 2020]

3.2 Geometric Stochastic Block Model (GSBM) (quid du GKBM?)

[Sankararaman and Baccelli, 2018] [Abbe et al., 2021] [Avrachenkov et al., 2024]

3.3 Geometric Hidden Community Model (GHCM)

[Gaudio et al., 2024b] [Gaudio et al., 2024a] [Gaudio and Guan, 2025] [Gaudio and Jin, 2026]

4 Spectral

(A confirmer) [Chen et al., 2016]
 (papier SBM mais théorie intéressante) [Bordenave et al., 2015]
 Cadre spectral intéressant pour certains espaces [Méliot, 2019]
 Méthode spectrale + Robustesse (la perturbation est géométrique) [Péché and Perchet, 2020]
 Méthode spectrale sur Soft Geometric Block Model (SGBM) (Generalisation de Geometric SBM) [Avrachenkov et al., 2021]

5 Ouvertures

- Considérer des modèles hyperboliques, pas seulement labels & distance-dependant, car ceux-ci captent mieux les propriétés habituelles des réseaux
 - Clustering [Krioukov, 2016]
 - Degré inhomogène [Barabási and Albert, 1999]
 - "Communities with specific mixing patterns" (mesoscale) [Toivonen et al., 2006]
- Communautés apparaissent dans un espace hyperbolique latent [Bruno et al., 2019] Réponse 6ans plus tard : pas toujours! + définition d'un modèle pour "régler" ça [Guarino et al., 2025] [Li et al., 2018]

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